基于 ARMA-EGARCH 模型的国际原油价格时间序列分析

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2026 年 4 月 26 日

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1 引言

国际原油价格波动深刻影响着全球经济运行、能源安全及金融市场稳定。WTI、Brent 和 Dubai 三大基准原油分别代表北美、欧洲和亚太三大区域市场，其价格波动特征既存在共性，又因市场结构差异而呈现不同特征。准确刻画原油价格的波动规律对投资决策、风险管理及政策制定具有重要意义。

传统时间序列模型通常假设扰动项同方差，但金融资产收益率序列普遍存在波动聚集和杠杆效应。自 Engle(1982) 提出 ARCH 模型以来，GARCH 族模型成为刻画金融波动的主流工具。其中，EGARCH 模型能够捕捉波动非对称性，且无需对参数施加非负约束，更适用于原油等大宗商品市场。

本文选取 1992 年 1 月至 2026 年 3 月 WTI、Brent 和 Dubai 原油月度价格数据，首先对对数收益率序列构建 ARMA 均值模型，然后基于 t 分布构建 EGARCH 族模型，系统对比三大原油的波动特征，为市场参与者提供量化参考。

2 数据来源与预处理

2.1 数据基本信息介绍

本文选取全球三大原油基准价格作为研究对象，分别为：西德克萨斯中质原油 (WTI)、布伦特原油 (Brent) 及迪拜原油 (Dubai)，其基本信息如下：

- 数据名称：

  Global price of WTI Crude (POILWTIUSDM)，

  Global price of Brent Crude (POILBREUSDM)，

  Global price of Dubai Crude (POILDUBUSDM)，

- 数据来源：

  https://fred.stlouisfed.org/series/POILWTIUSDM，

  https://fred.stlouisfed.org/series/POILBREUSDM，

  https://fred.stlouisfed.org/series/POILDUBUSDM，

- 数据频率：月度数据；时间跨度：1992 年 1 月—2026 年 3 月

- 分析软件：EViews 13

- 三大原油价格的基各项指标如表 1所示。

表 1: 三大原油基准价格各项指标说明

系列代号	名称	本质	产地	品质	定价基准
POILWTIUSDM	WTI	西德克萨斯中质原油	美国得州	轻质低硫	北美原油贸易基准
POILBREUSDM	Brent	北海布伦特原油	欧洲北海	轻质低硫	全球原油贸易核心基准
POILDUBUSDM	Dubai	迪拜原油	中东阿联酋	中质含硫	亚太原油进口基准

2.2　数据预处理

为消除量纲影响和平滑波动，分别对原始序列 POILWTIUSDM、POILBREUSDM、POILDUBUSDM 做对数变换得

$$\mathrm{LN_WTI} = \ln\left(\mathrm{POILWTIUSDM}\right), \tag{1}$$

$$\mathrm{LN_BRENT} = \ln\left(\mathrm{POILBREUSDM}\right), \tag{2}$$

$$\mathrm{LN_DUBAI} = \ln\left(\mathrm{POILDUBUSDM}\right), \tag{3}$$

其时序图如图 1所示。

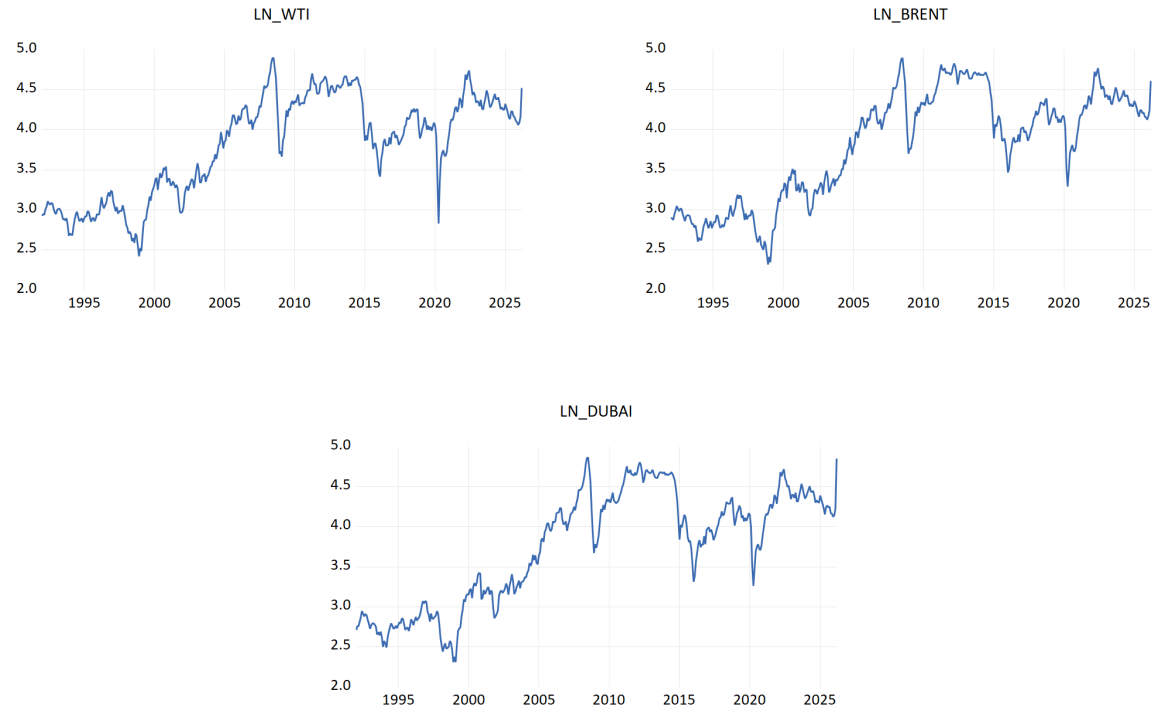


图 1: 三大原油价格对数序列时序图

3　构建序列 LN_WTI 的时间序列模型

3.1　平稳性与非白噪声检验

3.1.1　ADF 单位根检验

对序列 LN_WTI 做三种情形的 ADF 单位根检验，结果如图 2所示。

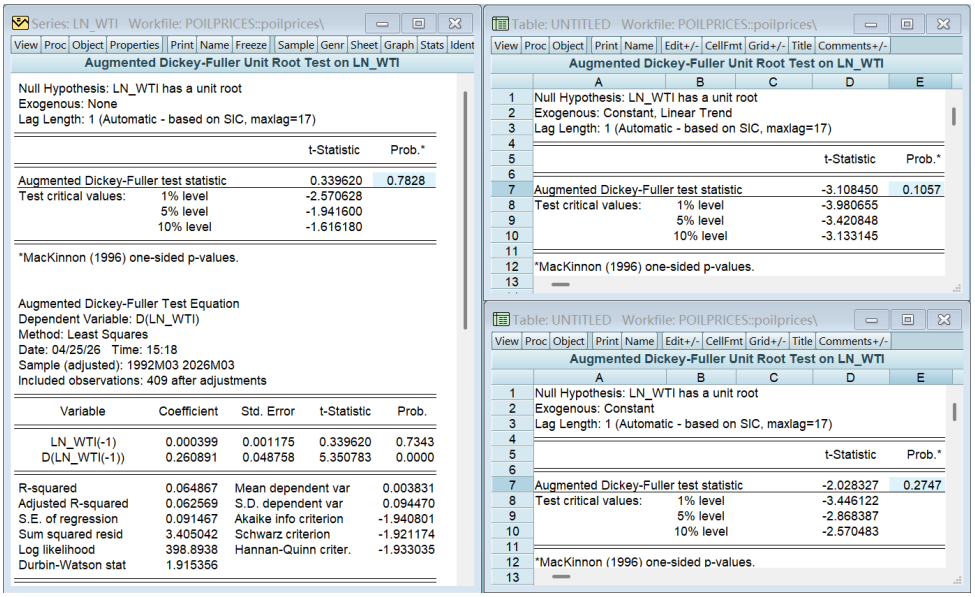


图 2: 序列 LN_WTI 的 ADF 检验结果图

由图 2可知，序列 LN_WTI 三种情形的 P 值均大于 0.05，在 5% 显著水平下均无法拒绝“序列存在单位根”的原假设。这说明，序列 LN_WTI 在三种检验设定下均存在单位根、不满足平稳性要求。因此，后续建模不能直接对原序列建立 ARMA 模型，需要先对序列进行差分处理，再对差分后的序列重新进行单位根检验，根据差分后的平稳性结果选择对应的模型形式。

对序列 LN_WTI 做一阶差分得序列

$$\mathrm{DF_LN_WTI} = \mathrm{LN_WTI}_t - \mathrm{LN_WTI}_{t-1}, \tag{4}$$

其时序图如图 3所示。

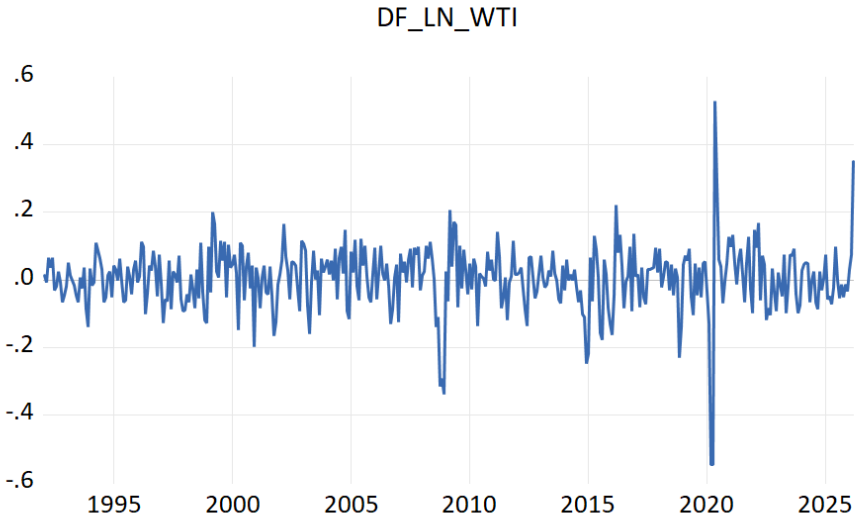
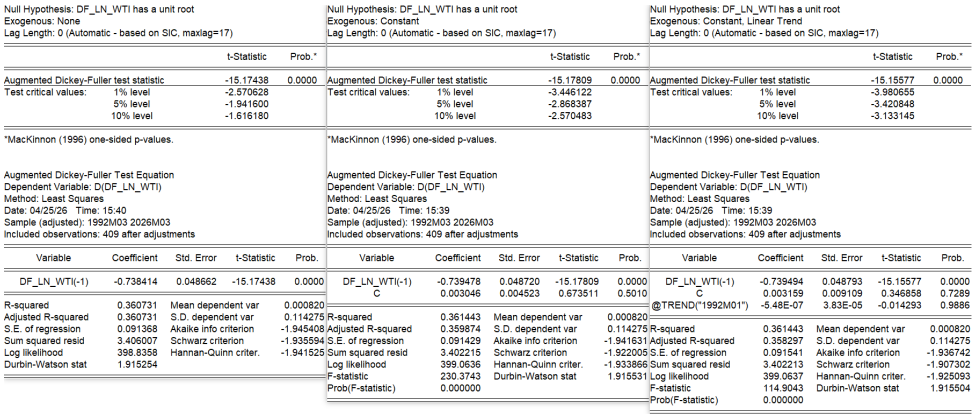


图 3: 序列 DF_LN_WTI 时序图

再对序列 DF_LN_WTI 做三种情形的 ADF 单位根检验，结果如图 4所示。



Null Hypothesis: DF_LN_WTI has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=17)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-15.17438	0.0000
Test critical values:	1% level	-2.570628	
	5% level	-1.941600	
	10% level	-1.616180	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DF_LN_WTI)
Method: Least Squares
Date: 04/25/26 Time: 15:40
Sample (adjusted): 1992M03 2026M03
Included observations: 409 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DF_LN_WTI(-1)	-0.738414	0.048662	-15.17438	0.0000

R-squared	0.360731	Mean dependent var	0.000820
Adjusted R-squared	0.360731	S.D. dependent var	0.114275
S.E. of regression	0.091368	Akaike info criterion	-1.945408
Sum squared resid	3.406007	Schwarz criterion	-1.935594
Log likelihood	398.8358	Hannan-Quinn criter.	-1.941525
Durbin-Watson stat	1.915254		

Null Hypothesis: DF_LN_WTI has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=17)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-15.17809	0.0000
Test critical values:	1% level	-3.446122	
	5% level	-2.868387	
	10% level	-2.570483	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DF_LN_WTI)
Method: Least Squares
Date: 04/25/26 Time: 15:39
Sample (adjusted): 1992M03 2026M03
Included observations: 409 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DF_LN_WTI(-1)	-0.739478	0.048720	-15.17809	0.0000
C	0.003046	0.004523	0.673511	0.5010

R-squared	0.361443	Mean dependent var	0.000820
Adjusted R-squared	0.359874	S.D. dependent var	0.114275
S.E. of regression	0.091429	Akaike info criterion	-1.941631
Sum squared resid	3.402215	Schwarz criterion	-1.922005
Log likelihood	399.0636	Hannan-Quinn criter.	-1.933866
F-statistic	230.3743	Durbin-Watson stat	1.915531
Prob(F-statistic)	0.000000		

Null Hypothesis: DF_LN_WTI has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=17)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-15.15577	0.0000
Test critical values:	1% level	-3.980655	
	5% level	-3.420848	
	10% level	-3.133145	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DF_LN_WTI)
Method: Least Squares
Date: 04/25/26 Time: 15:39
Sample (adjusted): 1992M03 2026M03
Included observations: 409 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DF_LN_WTI(-1)	-0.739494	0.048793	-15.15577	0.0000
C	0.003159	0.009109	0.346858	0.7289
@TREND("1992M01")	-5.48E-07	3.83E-05	-0.014293	0.9886

R-squared	0.361443	Mean dependent var	0.000820
Adjusted R-squared	0.358297	S.D. dependent var	0.114275
S.E. of regression	0.091541	Akaike info criterion	-1.936742
Sum squared resid	3.402213	Schwarz criterion	-1.907302
Log likelihood	399.0637	Hannan-Quinn criter.	-1.925093
F-statistic	114.9043	Durbin-Watson stat	1.915504
Prob(F-statistic)	0.000000		

图 4: 序列 DF_LN_WTI 的 ADF 检验结果图

由图 4可知，序列 DF_LN_WTI 三种情形的 p 值均小于 0.05，在 5% 显著性水平下均拒绝“序列存在单位根”的原假设。这说明，序列 DF_LN_WTI 在三种检验设定下均不存在单位根、满足平稳性要求。再将图 4中三种情形的 AIC/SC/HQ 值汇总到表 2中。

表 2: 序列 DF_LN_WTI 单位根检验不同情形的信息准则对比

检验设定	AIC 准则	SC 准则	HQ 准则
None	**−1.945408**	**−1.935594**	**−1.941525**
Constant	-1.941631	-1.922005	-1.933866
Constant,Linear Trend	-1.936742	-1.907302	-1.925093

根据 AIC/SC/HQ 准则，最终选择**无截距项、无时间趋势** (**None**) 的情形，故可对序列 DF_LN_WTI 构建**无截距、无时间趋势的 ARMA 模型**。

3.1.2 自相关与偏自相关函数分析

序列 DF_LN_WTI 的自相关与偏自相关函数检验结果如图 5所示。

Date: 04/25/26 Time: 15:40
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.252	0.252	26.220	0.000
		2	-0.031	-0.101	26.615	0.000
		3	-0.093	-0.064	30.194	0.000
		4	-0.092	-0.058	33.737	0.000
		5	-0.008	0.023	33.767	0.000
		6	-0.011	-0.031	33.818	0.000
		7	-0.022	-0.023	34.022	0.000
		8	-0.025	-0.020	34.282	0.000
		9	-0.025	-0.019	34.555	0.000
		10	0.015	0.020	34.651	0.000
		11	-0.001	-0.020	34.652	0.000
		12	-0.029	-0.030	34.999	0.000
		13	-0.061	-0.052	36.568	0.000
		14	-0.058	-0.035	38.019	0.001
		15	-0.016	-0.006	38.123	0.001
		16	0.049	0.039	39.135	0.001

图 5: 序列 DF_LN_WTI 的 ACF 和 PACF 相关图

由图 5可知

- 滞后 1 至 16 阶的 Q 统计量伴随概率 (p 值) 均小于 0.05 的显著性水平，因此拒绝“序列为白噪声”的原假设，表明序列 DF_LN_WTI 存在显著的自相关性，并非独立同分布的白噪声序列，具备构建时间序列模型的基础。
- 自相关函数 (ACF) 仅滞后 1 阶系数显著不为 0，滞后 2 阶及以后的系数绝对值均小于临界值，呈现 **1 阶截尾**特征；偏自相关函数 (PACF) 在滞后 1 阶、滞后 2 阶显著不为 0，自滞后 3 阶起所有系数均未通过显著性检验，呈现 **2 阶截尾**特征。

3.2 ARMA 模型构建与检验

3.2.1 模型构建与优选

根据之前对序列 DF_LN_WTI 的自相关函数 (ACF) 和偏自相关函数 (PACF) 的特征分析，构建 AR(1)、AR(2)、MA(1)、MA(2)、ARMA(1,1)、ARMA(1,2)、ARMA(2,1)、

ARMA(2,2) 模型，再残差检验和信息准则筛选最优模型。各个模型的结果如图 6和图 7所示。

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:38
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 68 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.260940	0.018958	13.76403	0.0000
SIGMASQ	0.008308	0.000258	32.24304	0.0000

R-squared	0.064578	Mean dependent var	0.003856
Adjusted R-squared	0.062285	S.D. dependent var	0.094356
S.E. of regression	0.091370	Akaike info criterion	-1.942758
Sum squared resid	3.406190	Schwarz criterion	-1.923167
Log likelihood	400.2653	Hannan-Quinn criter.	-1.935007
Durbin-Watson stat	1.914201		

Inverted AR Roots	.26

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:38
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 68 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.287350	0.033957	8.462254	0.0000
AR(2)	-0.103553	0.043353	-2.388590	0.0174
SIGMASQ	0.008221	0.000301	27.31300	0.0000

R-squared	0.074368	Mean dependent var	0.003856
Adjusted R-squared	0.069820	S.D. dependent var	0.094356
S.E. of regression	0.091002	Akaike info criterion	-1.948350
Sum squared resid	3.370539	Schwarz criterion	-1.918963
Log likelihood	402.4117	Hannan-Quinn criter.	-1.936723
Durbin-Watson stat	1.978661		

Inverted AR Roots	.14+.29i	.14-.29i

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:40
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 118 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.284941	0.031217	9.127804	0.0000
MA(2)	0.002880	0.031008	0.092890	0.9260
SIGMASQ	0.008246	0.000289	28.52278	0.0000

R-squared	0.071486	Mean dependent var	0.003856
Adjusted R-squared	0.066923	S.D. dependent var	0.094356
S.E. of regression	0.091144	Akaike info criterion	-1.945259
Sum squared resid	3.381034	Schwarz criterion	-1.915872
Log likelihood	401.7781	Hannan-Quinn criter.	-1.933633
Durbin-Watson stat	1.966330		

Inverted MA Roots	-.01	-.27

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:39
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 36 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.284026	0.026707	10.63474	0.0000
SIGMASQ	0.008246	0.000250	33.04331	0.0000

R-squared	0.071480	Mean dependent var	0.003856
Adjusted R-squared	0.069204	S.D. dependent var	0.094356
S.E. of regression	0.091032	Akaike info criterion	-1.950131
Sum squared resid	3.381057	Schwarz criterion	-1.930540
Log likelihood	401.7768	Hannan-Quinn criter.	-1.942380
Durbin-Watson stat	1.964787		

Inverted MA Roots	-.28

图 6: 序列 DF_LN_WTI 的 AR、MA 模型结果图

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:42
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 90 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.007248	0.109330	0.066291	0.9472
MA(1)	0.277438	0.128427	2.160274	0.0313
SIGMASQ	0.008246	0.000289	28.49288	0.0000

R-squared	0.071484	Mean dependent var	0.003856
Adjusted R-squared	0.066922	S.D. dependent var	0.094356
S.E. of regression	0.091144	Akaike info criterion	-1.945257
Sum squared resid	3.381041	Schwarz criterion	-1.915871
Log likelihood	401.7777	Hannan-Quinn criter.	-1.933631
Durbin-Watson stat	1.965897		

Inverted AR Roots	.01
Inverted MA Roots	-.28

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:42
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 111 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	1.038797	0.118265	8.783639	0.0000
AR(2)	-0.146959	0.129744	-1.132682	0.2580
MA(1)	-0.767212	0.138252	-5.549357	0.0000
MA(2)	-0.159573	0.142373	-1.120808	0.2630
SIGMASQ	0.008132	0.000309	26.30955	0.0000

R-squared	0.084403	Mean dependent var	0.003856
Adjusted R-squared	0.075360	S.D. dependent var	0.094356
S.E. of regression	0.090731	Akaike info criterion	-1.949199
Sum squared resid	3.334001	Schwarz criterion	-1.900222
Log likelihood	404.5859	Hannan-Quinn criter.	-1.929823
Durbin-Watson stat	1.965977		

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:42
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 97 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.978940	0.163813	5.975962	0.0000
AR(2)	-0.275565	0.048963	-5.628045	0.0000
MA(1)	-0.706221	0.175007	-4.035380	0.0001
SIGMASQ	0.008165	0.000299	27.31661	0.0000

R-squared	0.080618	Mean dependent var	0.003856
Adjusted R-squared	0.073824	S.D. dependent var	0.094356
S.E. of regression	0.090806	Akaike info criterion	-1.950169
Sum squared resid	3.347783	Schwarz criterion	-1.910987
Log likelihood	403.7846	Hannan-Quinn criter.	-1.934667
Durbin-Watson stat	1.959620		

Inverted AR Roots	.49-.19i	.49+.19i
Inverted MA Roots	.71	

Dependent Variable: DF_LN_WTI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/25/26 Time: 16:41
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 79 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.897061	0.056923	15.75920	0.0000
MA(1)	-0.636983	0.075296	-8.459694	0.0000
MA(2)	-0.293868	0.041217	-7.129855	0.0000
SIGMASQ	0.008148	0.000305	26.71026	0.0000

R-squared	0.082592	Mean dependent var	0.003856
Adjusted R-squared	0.075813	S.D. dependent var	0.094356
S.E. of regression	0.090709	Akaike info criterion	-1.952144
Sum squared resid	3.340592	Schwarz criterion	-1.912962
Log likelihood	404.1895	Hannan-Quinn criter.	-1.936642
Durbin-Watson stat	1.943970		

Inverted AR Roots	.90	
Inverted MA Roots	.95	-.31

图 7: 序列 DF_LN_WTI 的 ARMA 模型结果图

将图 6和图 7中各个模型的系数显著性检验结果和 AIC/SC/HQ 值分别汇总到如下表 3和表 4中。

表 3: ARMA 模型系数显著性检验结果

模型	AR(1)	AR(2)	MA(1)	MA(2)
ARMA(1,1)	0.9472	-	0.0313	-
ARMA(2,2)	0.0000	0.2580	0.0000	0.2630
ARMA(2,1)	0.0000	0.0000	0.0001	-
ARMA(1,2)	0.0000	-	0.0000	0.0000
AR(1)	0.0000	-	-	-
AR(2)	0.0000	0.0174	-	-
MA(1)	-	-	0.0000	-
MA(2)	-	-	0.0000	0.9260

表 4: ARMA 模型信息准则对比结果

模型	AIC	SC	HQ
ARMA(1,1)	-1.9453	-1.9159	-1.9336
ARMA(2,2)	-1.9492	-1.9002	-1.9298
ARMA(2,1)	-1.9502	-1.9109	-1.9347
ARMA(1,2)	-1.9521	-1.9130	-1.9366
AR(1)	-1.9428	-1.9232	-1.9350
AR(2)	-1.9483	-1.9189	-1.9367
MA(1)	**−1.9501**	**−1.9305**	**−1.9424**
MA(2)	-1.9453	-1.9159	-1.9336

根据系数显著性和 AIC/SC/HQ 信息准则，再结合表 3和表 4，最终选择 **MA(1) 模型**进行后续建模与分析。其余模型或存在参数不显著的问题，或在信息准则上表现略逊，因此不予采用。

3.2.2 适应性检验

对序列 DF_LN_WTI 的 MA(1) 模型的残差序列做 Q 检验，结果如图 8所示。

Date: 04/25/26 Time: 19:45
Sample (adjusted): 1992M02 2026M03
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.000	0.000	6.E-05	
		2	-0.011	-0.011	0.0519	0.820
		3	-0.068	-0.068	1.9820	0.371
		4	-0.079	-0.080	4.5838	0.205
		5	0.015	0.013	4.6801	0.322
		6	-0.012	-0.019	4.7445	0.448
		7	-0.015	-0.026	4.8421	0.564
		8	-0.015	-0.020	4.9339	0.668
		9	-0.028	-0.029	5.2756	0.728
		10	0.026	0.020	5.5609	0.783
		11	-0.004	-0.010	5.5673	0.850
		12	-0.017	-0.024	5.6956	0.893
		13	-0.046	-0.048	6.5843	0.884
		14	-0.044	-0.044	7.4172	0.879
		15	-0.013	-0.022	7.4933	0.914
		16	0.043	0.031	8.2752	0.912

图 8: 序列 DF_LN_WTI 的 MA(1) 模型的适应性检验结果图

由图 8可知

- 滞后 2 至 16 阶的 Q 统计量伴随概率均大于 0.05 的显著性水平，无法拒绝“残差序列为白噪声”的原假设。
- 残差序列的自相关函数 (ACF) 与偏自相关函数 (PACF) 在各滞后阶数的系数绝对值均未超出 95% 置信区间，无显著自相关性，表明 MA(1) 模型已充分提取原序列的均值自相关信息。

3.2.3 条件异方差检验

考虑到条件异方差多表现为短期波动聚集，低阶滞后项已能捕捉绝大多数 ARCH 效应；同时，滞后阶数过高会导致检验自由度下降、估计偏差增大，故本文只选取滞后 4、5、6 阶进行 ARCH-LM 检验。现在对序列 DF_LN_WTI 的 MA(1) 模型的残差分别做滞后 4、5、6 阶的 ARCH 效应检验，结果如图 9所示。

Heteroskedasticity Test: ARCH

F-statistic	35.38675	Prob. F(4,401)	0.0000
Obs*R-squared	105.9228	Prob. Chi-Square(4)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/25/26 Time: 19:23
Sample (adjusted): 1992M06 2026M03
Included observations: 406 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005080	0.001277	3.978229	0.0001
RESID^2(-1)	0.333560	0.051074	6.530921	0.0000
RESID^2(-2)	0.372988	0.052479	7.107375	0.0000
RESID^2(-3)	-0.238260	0.052478	-4.540165	0.0000
RESID^2(-4)	-0.066428	0.051071	-1.300681	0.1941

R-squared	0.260894	Mean dependent var	0.008314
Adjusted R-squared	0.253521	S.D. dependent var	0.027291
S.E. of regression	0.023580	Akaike info criterion	-4.644636
Sum squared resid	0.222954	Schwarz criterion	-4.595297
Log likelihood	947.8612	Hannan-Quinn criter.	-4.625109
F-statistic	35.38675	Durbin-Watson stat	1.932781
Prob(F-statistic)	0.000000		

Heteroskedasticity Test: ARCH

F-statistic	30.63522	Prob. F(5,399)	0.0000
Obs*R-squared	112.3488	Prob. Chi-Square(5)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/25/26 Time: 19:24
Sample (adjusted): 1992M07 2026M03
Included observations: 405 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.004357	0.001291	3.376281	0.0008
RESID^2(-1)	0.343313	0.050734	6.766954	0.0000
RESID^2(-2)	0.409384	0.053392	7.667537	0.0000
RESID^2(-3)	-0.295577	0.055365	-5.338679	0.0000
RESID^2(-4)	-0.117755	0.053392	-2.205503	0.0280
RESID^2(-5)	0.153252	0.050730	3.020967	0.0027

R-squared	0.277404	Mean dependent var	0.008326
Adjusted R-squared	0.268349	S.D. dependent var	0.027324
S.E. of regression	0.023372	Akaike info criterion	-4.659837
Sum squared resid	0.217957	Schwarz criterion	-4.600520
Log likelihood	949.6170	Hannan-Quinn criter.	-4.636358
F-statistic	30.63522	Durbin-Watson stat	1.939793
Prob(F-statistic)	0.000000		

Heteroskedasticity Test: ARCH

F-statistic	25.51322	Prob. F(6,397)	0.0000
Obs*R-squared	112.4275	Prob. Chi-Square(6)	0.0000

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/25/26 Time: 19:24
Sample (adjusted): 1992M08 2026M03
Included observations: 404 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.004515	0.001311	3.442611	0.0006
RESID^2(-1)	0.349045	0.051444	6.784906	0.0000
RESID^2(-2)	0.405008	0.053824	7.524710	0.0000
RESID^2(-3)	-0.306544	0.057505	-5.330706	0.0000
RESID^2(-4)	-0.102463	0.057505	-1.781800	0.0755
RESID^2(-5)	0.166044	0.053823	3.085010	0.0022
RESID^2(-6)	-0.037345	0.051439	-0.725995	0.4683

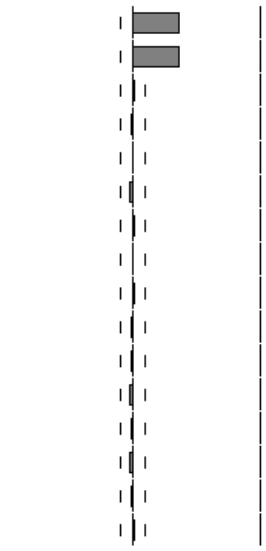
R-squared	0.278286	Mean dependent var	0.008341
Adjusted R-squared	0.267378	S.D. dependent var	0.027356
S.E. of regression	0.023415	Akaike info criterion	-4.653689
Sum squared resid	0.217664	Schwarz criterion	-4.584357
Log likelihood	947.0451	Hannan-Quinn criter.	-4.626243
F-statistic	25.51322	Durbin-Watson stat	1.953883
Prob(F-statistic)	0.000000		

图 9: MA(1) 模型的 ARCH 效应检验

图 9的结果显示：在三种滞后阶数下，F 统计量与 LM 统计量的 p 值均显著为 0.0000<0.05，表明残差平方序列存在显著的条件异方差性。

再对序列的残差平方做检验，结果如图 10所示。

Date: 04/25/26 Time: 19:35
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.372	0.372	57.070	0.000
		2	0.371	0.270	114.06	0.000
		3	0.009	-0.240	114.09	0.000
		4	-0.010	-0.064	114.13	0.000
		5	-0.004	0.135	114.14	0.000
		6	-0.020	-0.030	114.30	0.000
		7	0.011	-0.034	114.35	0.000
		8	-0.006	0.019	114.37	0.000
		9	0.012	0.018	114.43	0.000
		10	-0.010	-0.029	114.48	0.000
		11	-0.016	-0.020	114.59	0.000
		12	-0.025	0.001	114.85	0.000
		13	-0.016	0.007	114.96	0.000
		14	-0.025	-0.025	115.22	0.000
		15	-0.012	-0.001	115.28	0.000
		16	0.013	0.043	115.35	0.000

图 10: MA(1) 模型残差的平方的 Q 检验结果图

由图 10可知

- 滞后 1 至 16 阶的 Q 统计量的伴随概率均小于 0.01，在 1% 的显著性水平下强烈拒绝“残差平方序列不存在自相关”的原假设。
- 残差平方序列的自相关函数 (ACF) 在滞后 1 阶 (系数为 0.372)、滞后 2 阶 (系数为 0.371) 均显著不为 0，呈现出明显的自相关性，表明残差序列存在显著的波动率聚集性特征。

上述两种检验的结果说明，序列 DF_LN_WTI 的 MA(1) 模型的残差序列存在显著的条件异方差，不满足经典时间序列模型的同方差假设。为此，本文进一步构建 GARCH 类模型，以同时刻画序列的均值过程与异方差特征。

3.3 GARCH 族模型构建与检验

序列 DF_LN_WTI 的直方图与描述性统计如图 11所示。

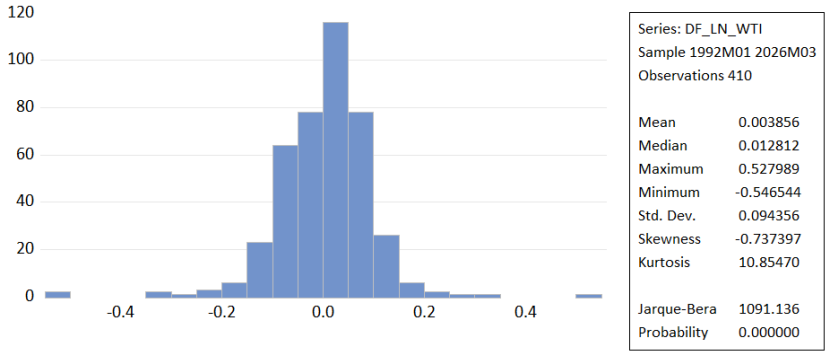


图 11: 序列 DF_LN_WTI 的直方图与描述性统计图

由图 11可知，序列 DF_LN_WTI 的峰度为 10.85，显著大于正态分布的峰度 3，且 Jarque-Bera 正态性检验的伴随概率为 0.0000，强烈拒绝序列服从正态分布的原假设，表明该序列具有典型的**尖峰厚尾**特征。传统 GARCH 模型的正态分布假设难以刻画金融时间序列的厚尾性，而 t 分布能够更好地拟合极端波动的尾部风险。基于此，本文后续构建的 GARCH 模型采用 **t 分布**设定扰动项分布，以提升模型对序列波动特征的拟合效果。

3.3.1　模型构建与优选

基于序列 DF_LN_WTI 的尖峰厚尾特征与显著的条件异方差效应，对序列构建 GARCH 族模型，以估计序列的波动特征。

由于金融时间序列的异方差与波动持续性主要体现于短期滞后项，低阶 GARCH 模型即可有效刻画序列波动特征。过高滞后阶数易造成待估参数增多、拟合过度、参数估计不稳定等问题。因此本文在 MA(1) 均值方程基础上，只选取 MA(1)-GARCH、MA(1)-TARCH、MA(1)-EGARCH 三类低阶模型，并采用 Student's t 分布以适配序列 DF_LN_WTI 尖峰厚尾的统计特征。将各 GARCH 族模型系数显著性检验和信息准则结果分别汇总到表 5和表 6中。

表 5: DF_LN_WTI 序列 GARCH 族模型系数显著性检验结果

模型名称	常数项 C	ARCH 项	非对称项	GARCH 项
MA(1)-GARCH(1,0)-t	**0.0000**	**0.0033**	-	-
MA(1)-GARCH(2,0)-t	0.0000	0.0400/0.0504	-	-
MA(1)-GARCH(1,1)-t	**0.0292**	**0.0075**	-	**0.0002**
MA(1)-GARCH(1,2)-t	0.0514	0.0280	-	0.2649/0.9921
MA(1)-GARCH(2,1)-t	**0.0617**	**0.0046/0.0042**	-	**0.0000**
MA(1)-GARCH(2,2)-t	0.1663/0.0122	0.0116	-	0.0000/0.0211
MA(1)-TARCH(1,0)-t	0.0000	0.7631	0.0012	-
MA(1)-TARCH(2,0)-t	0.0000	0.7519/0.1519	0.0017	-
MA(1)-TARCH(1,1)-t	0.0000	0.8927	0.0014	0.7877
MA(1)-TARCH(1,2)-t	0.0000	0.9659	0.0004	0.0935/0.0513
MA(1)-TARCH(2,1)-t	0.0000	0.9134/0.1907	0.0012	0.5160
MA(1)-TARCH(2,2)-t	0.0000	0.8634/0.3143	0.0011	0.3208/0.4738
MA(1)-EGARCH(1,0)-t	0.0000	0.0625	0.0016	-
MA(1)-EGARCH(2,0)-t	0.0000	0.1981/0.0000	0.0020	-
MA(1)-EGARCH(1,1)-t	**0.0113**	**0.0158**	**0.0193**	**0.0000**
MA(1)-EGARCH(1,2)-t	0.0232	0.0381	0.0121	0.0293/0.0797
MA(1)-EGARCH(2,1)-t	0.0001	0.1802/0.0042	0.0028	0.6293
MA(1)-EGARCH(2,2)-t	0.0104	0.1807/0.0272	0.0037	0.4647/0.0447

表 6: DF_LN_WTI 序列 GARCH 族模型信息准则结果对比

模型名称	MA(1)	AIC	SC	HQ
MA(1)-GARCH(1,0)-t	0.0030	-2.1922	-2.1531	-2.1767
MA(1)-GARCH(2,0)-t	0.0001	-2.2011	-2.1520	-2.1817
MA(1)-GARCH(1,1)-t	0.0008	-2.2037	-2.1547	-2.1843
MA(1)-GARCH(1,2)-t	0.0008	-2.1988	-2.1400	-2.1755
MA(1)-GARCH(2,1)-t	0.0016	-2.2082	-2.1495	-2.1850
MA(1)-GARCH(2,2)-t	0.0005	-2.2140	-2.1418	-2.1832
MA(1)-TARCH(1,0)-t	0.0000	-2.2342	-2.1852	-2.2148
MA(1)-TARCH(2,0)-t	0.0000	-2.2377	-2.1789	-2.2144
MA(1)-TARCH(1,1)-t	0.0000	-2.2296	-2.1708	-2.2063
MA(1)-TARCH(1,2)-t	0.0000	-2.2323	-2.1637	-2.2052
MA(1)-TARCH(2,1)-t	0.0000	-2.2357	-2.1671	-2.2086
MA(1)-TARCH(2,2)-t	0.0000	-2.2332	-2.1548	-2.2022
MA(1)-EGARCH(1,0)-t	0.0000	-2.1906	-2.1417	-2.1713
MA(1)-EGARCH(2,0)-t	0.0000	-2.2231	-2.1644	-2.1999
MA(1)-EGARCH(1,1)-t	0.0000	**-2.2182**	**-2.1594**	**-2.1949**
MA(1)-EGARCH(1,2)-t	0.0000	-2.2155	-2.1469	-2.1883
MA(1)-EGARCH(2,1)-t	0.0000	-2.2187	-2.1501	-2.1915
MA(1)-EGARCH(2,2)-t	0.0000	-2.2252	-2.1468	-2.1942

由表 5和表 6，再结合显著性检验和信息准则可知，$\mathbf{MA(1)-EGARCH(1,1)-t}$ **模型**不仅所有参数均通过显著性检验，有效刻画了序列的杠杆效应与波动持续性，且信息准则表现最优，是刻画序列 DF_LN_WTI 波动特征的最优模型，其结果如图 12所示。

Dependent Variable: DF_LN_WTI
Method: ML ARCH - Student's t distribution (OPG - BHHH / Marquardt steps)
Date: 04/25/26 Time: 20:41
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments
Convergence achieved after 46 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1992M01
Presample variance: backcast (parameter = 0.7)
LOG(GARCH) = C(2) + C(3)*ABS(RESID(-1)/@SQRT(GARCH(-1))) + C(4)*RESID(-1)/@SQRT(GARCH(-1)) + C(5)*LOG(GARCH(-1))

Variable	Coefficient	Std. Error	z-Statistic	Prob.
MA(1)	0.227205	0.054176	4.193873	0.0000
Variance Equation				
C(2)	-1.250255	0.493649	-2.532680	0.0113
C(3)	0.297440	0.123248	2.413338	0.0158
C(4)	-0.162100	0.069255	-2.340636	0.0193
C(5)	0.797172	0.088665	8.990818	0.0000
T-DIST. DOF	9.407736	2.742039	3.430928	0.0006

R-squared	0.068299	Mean dependent var	0.003856
Adjusted R-squared	0.068299	S.D. dependent var	0.094356
S.E. of regression	0.091077	Akaike info criterion	-2.218165
Sum squared resid	3.392640	Schwarz criterion	-2.159392
Log likelihood	460.7238	Hannan-Quinn criter.	-2.194913
Durbin-Watson stat	1.856121		

Inverted MA Roots	-.23

图 12: MA(1)-EGARCH(1,1)-t 模型估计结果图

3.3.2 残差白噪声检验

对序列 DF_LN_WTI 的 MA(1)-EGARCH(1,1)-t 模型的残差序列做 Q 检验，结果如图 13所示。

Date: 04/25/26 Time: 21:26
Sample (adjusted): 1992M02 2026M03
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.043	0.043	0.7472	
		2	0.012	0.011	0.8101	0.368
		3	-0.040	-0.041	1.4761	0.478
		4	-0.039	-0.036	2.1088	0.550
		5	0.009	0.013	2.1436	0.709
		6	-0.020	-0.022	2.3132	0.804
		7	-0.031	-0.032	2.7043	0.845
		8	-0.018	-0.015	2.8381	0.900
		9	-0.010	-0.008	2.8768	0.942
		10	0.008	0.005	2.9052	0.968
		11	0.019	0.016	3.0639	0.980
		12	-0.019	-0.022	3.2156	0.988
		13	-0.083	-0.083	6.1291	0.909
		14	-0.053	-0.046	7.3175	0.885
		15	-0.015	-0.011	7.4083	0.918
		16	0.015	0.008	7.5042	0.942

*Probabilities may not be valid for this equation specification.

图 13: MA(1)-EGARCH(1,1)-t 模型残差序列的 Q 检验结果图

由图 13可知，滞后 1 至 16 阶的自相关系数 (AC) 与偏自相关系数 (PAC) 均未超出 95% 置信区间，且各阶 Q 统计量对应的伴随概率均大于 0.05 的显著性水平，无法拒绝残差序列为白噪声的原假设。这表明 MA(1)-EGARCH(1,1)-t 模型已充分提取了序列的均值与波动信息，残差序列不再存在显著的自相关性，模型设定合理。

3.3.3 条件异方差检验

对序列 DF_LN_WTI 的 MA(1)-EGARCH(1,1)-t 模型的残差序列分别做滞后 4、5、6 阶的 ARCH 效应检验，结果如图 14所示。

Heteroskedasticity Test: ARCH

F-statistic	1.189628	Prob. F(4,401)	0.3148
Obs*R-squared	4.761345	Prob. Chi-Square(4)	0.3127

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/25/26 Time: 21:39
Sample (adjusted): 1992M06 2026M03
Included observations: 406 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.024193	0.153415	6.675946	0.0000
WGT_RESID^2(-1)	0.033292	0.059868	0.556092	0.5785
WGT_RESID^2(-2)	0.093991	0.059787	1.572107	0.1167
WGT_RESID^2(-3)	-0.061605	0.059782	-1.030492	0.3034
WGT_RESID^2(-4)	-0.066389	0.059814	-1.109938	0.2677

R-squared	0.011727	Mean dependent var	1.023501
Adjusted R-squared	0.001869	S.D. dependent var	2.197082
S.E. of regression	2.195027	Akaike info criterion	4.422505
Sum squared resid	1932.076	Schwarz criterion	4.471844
Log likelihood	-892.7685	Hannan-Quinn criter.	4.442032
F-statistic	1.189628	Durbin-Watson stat	1.692079
Prob(F-statistic)	0.314779		

Heteroskedasticity Test: ARCH

F-statistic	0.987999	Prob. F(5,399)	0.4247
Obs*R-squared	4.952958	Prob. Chi-Square(5)	0.4216

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/25/26 Time: 21:39
Sample (adjusted): 1992M07 2026M03
Included observations: 405 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.997416	0.163782	6.089913	0.0000
WGT_RESID^2(-1)	0.034800	0.060092	0.579111	0.5628
WGT_RESID^2(-2)	0.095241	0.060034	1.586460	0.1134
WGT_RESID^2(-3)	-0.064449	0.060250	-1.069684	0.2854
WGT_RESID^2(-4)	-0.067183	0.059998	-1.119746	0.2635
WGT_RESID^2(-5)	0.028283	0.060040	0.471071	0.6378

R-squared	0.012230	Mean dependent var	1.022684
Adjusted R-squared	-0.000149	S.D. dependent var	2.199737
S.E. of regression	2.199901	Akaike info criterion	4.429406
Sum squared resid	1930.986	Schwarz criterion	4.488722
Log likelihood	-890.9546	Hannan-Quinn criter.	4.452884
F-statistic	0.987999	Durbin-Watson stat	1.691598
Prob(F-statistic)	0.424721		

Heteroskedasticity Test: ARCH

F-statistic	1.030912	Prob. F(6,397)	0.4046
Obs*R-squared	6.197968	Prob. Chi-Square(6)	0.4014

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/25/26 Time: 21:40
Sample (adjusted): 1992M08 2026M03
Included observations: 404 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.059377	0.172782	6.131287	0.0000
WGT_RESID^2(-1)	0.037259	0.060184	0.619080	0.5362
WGT_RESID^2(-2)	0.091551	0.060183	1.521212	0.1290
WGT_RESID^2(-3)	-0.067717	0.060429	-1.120618	0.2631
WGT_RESID^2(-4)	-0.060258	0.060397	-0.997691	0.3190
WGT_RESID^2(-5)	0.030295	0.060149	0.503671	0.6148
WGT_RESID^2(-6)	-0.067759	0.060136	-1.126765	0.2605

R-squared	0.015342	Mean dependent var	1.023682
Adjusted R-squared	0.000460	S.D. dependent var	2.202373
S.E. of regression	2.201866	Akaike info criterion	4.433663
Sum squared resid	1924.742	Schwarz criterion	4.502994
Log likelihood	-888.5999	Hannan-Quinn criter.	4.461108
F-statistic	1.030912	Durbin-Watson stat	1.695054
Prob(F-statistic)	0.404570		

图 14: MA(1)-EGARCH(1,1)-t 模型的 ARCH 效应检验结果图

由图 14可知，在滞后 4、5、6 阶条件下，ARCH-LM 检验的伴随概率均远大于 0.05 的显著性水平，无法拒绝“残差不存在 ARCH 效应”的原假设。这表明 MA(1)-

EGARCH(1,1)-t 模型已充分提取序列 DF_LN_WTI 的波动信息，残差序列不存在显著的条件异方差，模型设定合理且有效。

3.4 模型方程与结果解读

3.4.1 模型方程与结果解读

根据图 12可知，序列 DF_LN_WTI 的 MA(1)-EGARCH(1,1)-t 模型的均值方程与方差方程形式如下：

$$\mathrm{DF_LN_WTI}_t = \varepsilon_t + \theta\varepsilon_{t-1} = \varepsilon_t + 0.227205 \cdot \varepsilon_{t-1}, \tag{5}$$

$$\begin{aligned} \ln\left(\sigma_t^2\right) &= \omega + \alpha\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \gamma\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \beta\ln\left(\sigma_{t-1}^2\right) \\ &= -1.250255 + 0.297440 \cdot \left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| - 0.162100 \cdot \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \\ &\quad + 0.797172 \cdot \ln\left(\sigma_{t-1}^2\right), \end{aligned} \tag{6}$$

其中，ε_t 为扰动项，σ_t^2 为条件方差，θ 为移动平均系数，ω 为方差方程常数项，α 为 ARCH 项系数，γ 为非对称项系数，β 为 GARCH 项系数，模型扰动项 $\varepsilon_t \sim t(9.407736)$。

均值方程(5)中，MA(1) 系数 $\theta = 0.2272$ 在 5% 水平下显著为正。其中 DF_LN_WTI 为全球 WTI 原油现货价格的月度对数收益率，该结果表明 WTI 原油月度收益率存在显著的一阶移动平均特征，当期原油价格波动受前一期随机扰动的正向影响，原油价格波动具备短期记忆性特征。

由方差方程(6)可知

- ARCH 项系数 $\alpha = 0.2974$ 显著为正，说明地缘政治冲突、原油供需格局变化、全球宏观经济波动等外部信息冲击，会显著加剧 WTI 原油现货价格的月度波动幅度。
- GARCH 项系数 $\beta = 0.7972$ 显著为正且接近 1，体现出 WTI 原油价格波动具有极强的持续性，前期原油价格的大幅波动会持续传导至后续时期，符合国际原油市场波动聚集的典型特征。
- 非对称项系数 $\gamma = -0.1621$ 显著为负，证实 WTI 原油收益率序列存在显著**杠杆效应**，即原油需求下滑、地缘利空消息等负向冲击对价格波动的放大作用，显著强于供给收紧、经济利好等正向冲击。

4 构建序列 LN_BRENT 的时间序列模型

4.1 平稳性与非白噪声检验

4.1.1 ADF 单位根检验

对序列 LN_BRENT 做 ADF 检验得到三种情形的 p 值分别为：None(0.8150)、Constant(0.4108)、Constant,Linear Trend(0.1999) 均大于 0.05，故原序列存在单位根，不满足平稳性。对其进行一阶差分得到序列

$$\text{DF_LN_BRENT} = \text{LN_BRENT}_t - \text{LN_BRENT}_{t-1}, \tag{7}$$

再对差分后的序列重新进行 ADF 检验，检验结果如表 7所示。

表 7: 序列 DF_LN_BRENT 的 ADF 检验不同情形的信息准则对比

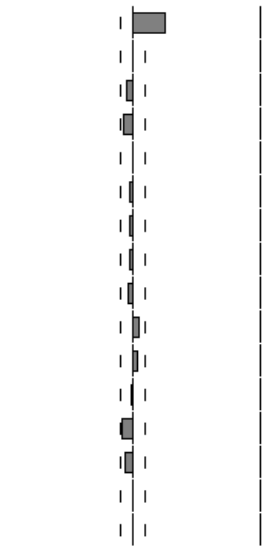
检验情形	p 值	AIC 准则	SC 准则	HQ 准则
None	0.0000	**−2.098702**	**−2.088888**	**−2.094819**
Constant	0.0000	-2.0953098	-2.075682	-2.087544
Constant,Linear Trend	0.0000	-2.090420	-2.060979	-2.078771

由表 7可知，三种检验情形下 p 值均小于 0.05，拒绝存在单位根的原假设，序列平稳。根据 AIC/SC/HQ 信息准则，选择**无截距项、无时间趋势**的情形进行建模。

4.1.2 自相关与偏自相关函数分析

序列 DF_LN_BRENT 自相关与偏自相关函数检验结果如图 15所示。

Date: 04/26/26 Time: 11:07
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.261	0.261	28.060	0.000
		2	0.005	-0.067	28.071	0.000
		3	-0.042	-0.028	28.810	0.000
		4	-0.075	-0.060	31.142	0.000
		5	-0.007	0.029	31.160	0.000
		6	-0.028	-0.042	31.488	0.000
		7	-0.023	-0.009	31.703	0.000
		8	-0.031	-0.030	32.119	0.000
		9	-0.039	-0.025	32.747	0.000
		10	0.042	0.057	33.497	0.000
		11	0.035	0.005	34.020	0.000
		12	-0.013	-0.030	34.088	0.001
		13	-0.085	-0.081	37.162	0.000
		14	-0.061	-0.012	38.766	0.000
		15	0.006	0.021	38.780	0.001
		16	0.005	-0.011	38.792	0.001

图 15: 序列 DF_LN_BRENT 的 ACF 和 PACF 相关图

由图 15可知，序列 DF_LN_BRENT 为非白噪声序列，其**自相关函数 1 阶截尾，偏自相关函数 1 阶截尾**。

4.2 ARMA 模型构建与条件异方差检验

4.2.1 模型构建与优选

根据序列 DF_LN_BRENT 自相关与偏自相关函数的特征，对 DF_LN_BRENT 构建多种 ARMA 模型，并将各个 ARMA 模型的系数显著性检验结果和 AIC/SC/HQ 值汇总到如下表 8中。

表 8: DF_LN_BRENT 序列 ARMA 模型系数显著性检验和信息准则对比图

模型	AR(1)	AR(2)	MA(1)	MA(2)	AIC	SC	HQ
ARMA(1,1)	0.6565	-	0.1044	-	-2.0954	-2.0660	-2.0837
ARMA(2,2)	0.0000	0.2551	0.0000	0.3959	-2.0928	-2.0438	-2.0734
ARMA(2,1)	0.0000	0.0000	0.0000	-	-2.0963	-2.0571	-2.0808
ARMA(1,2)	0.9917	-	0.9190	0.9894	-2.0905	-2.0513	-2.0750
AR(1)	0.0000	-	-	-	-2.0961	-2.0765	-2.0883
AR(2)	0.0000	0.0953	-	-	-2.0959	-2.0665	-2.0843
MA(1)	-	-	0.0000	-	**−2.1000**	**−2.0804**	**−2.0922**
MA(2)	-	-	0.0000	0.6528	-2.0954	-2.0660	-2.0837

由表 8，再结合系数显著性与信息准则，最优均值模型为 **MA(1)**，模型结果如图 16所示。

Dependent Variable: DF_LN_BRENT
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/26/26 Time: 11:45
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.286081	0.040862	7.001128	0.0000
SIGMASQ	0.007099	0.000333	21.32072	0.0000

R-squared	0.073062	Mean dependent var	0.004147
Adjusted R-squared	0.070790	S.D. dependent var	0.087618
S.E. of regression	0.084460	Akaike info criterion	-2.099999
Sum squared resid	2.910476	Schwarz criterion	-2.080408
Log likelihood	432.4998	Hannan-Quinn criter.	-2.092248
Durbin-Watson stat	1.950117		

Inverted MA Roots	-.29

图 16: 序列 DF_LN_BRENT 的 MA(1) 模型结果图

4.2.2　适应性与条件异方差检验

序列 DF_LN_BRENT 的 MA(1) 模型的适应性检验结果如图 17所示。

Date: 04/26/26　Time: 12:45
Sample (adjusted): 1992M02 2026M03
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.003	0.003	0.0050	
		2	0.013	0.013	0.0740	0.786
		3	-0.024	-0.025	0.3213	0.852
		4	-0.075	-0.075	2.6687	0.446
		5	0.022	0.023	2.8687	0.580
		6	-0.033	-0.032	3.3287	0.649
		7	-0.009	-0.013	3.3638	0.762
		8	-0.017	-0.021	3.4918	0.836
		9	-0.048	-0.046	4.4603	0.813
		10	0.052	0.047	5.6028	0.779
		11	0.020	0.020	5.7740	0.834
		12	-0.001	-0.008	5.7743	0.888
		13	-0.073	-0.079	8.0605	0.780
		14	-0.049	-0.040	9.0787	0.767
		15	0.021	0.022	9.2749	0.813
		16	-0.003	-0.006	9.2795	0.862

图 17: 序列 DF_LN_BRENT 的 MA(1) 模型适应性检验图

图 17显示：MA(1) 模型残差通过白噪声检验，均值信息提取充分。

序列 DF_LN_BRENT 的 MA(1) 模型的滞后 4、5、6 阶的 ARCH-LM 检验和残差平方序列的 Q 检验结果如图 18所示。

Heteroskedasticity Test: ARCH

F-statistic	4.702192	Prob. F(4,401)	0.0010
Obs*R-squared	18.19009	Prob. Chi-Square(4)	0.0011

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/26/26 Time: 12:58
Sample (adjusted): 1992M06 2026M03
Included observations: 406 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005325	0.000957	5.565927	0.0000
RESID^2(-1)	0.144909	0.053652	2.700914	0.0072
RESID^2(-2)	0.148079	0.054220	2.731092	0.0066
RESID^2(-3)	0.019102	0.054215	0.352335	0.7248
RESID^2(-4)	-0.046371	0.053643	-0.864434	0.3879

R-squared	0.044803	Mean dependent var	0.007149
Adjusted R-squared	0.035275	S.D. dependent var	0.015428
S.E. of regression	0.015154	Akaike info criterion	-5.528920
Sum squared resid	0.092082	Schwarz criterion	-5.479581
Log likelihood	1127.371	Hannan-Quinn criter.	-5.509393
F-statistic	4.702192	Durbin-Watson stat	1.866377
Prob(F-statistic)	0.001018		

Heteroskedasticity Test: ARCH

F-statistic	3.747387	Prob. F(5,399)	0.0025
Obs*R-squared	18.16564	Prob. Chi-Square(5)	0.0027

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/26/26 Time: 12:58
Sample (adjusted): 1992M07 2026M03
Included observations: 405 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005394	0.000996	5.413335	0.0000
RESID^2(-1)	0.144227	0.053829	2.679362	0.0077
RESID^2(-2)	0.148471	0.054367	2.730937	0.0066
RESID^2(-3)	0.020585	0.054951	0.374599	0.7082
RESID^2(-4)	-0.044985	0.054364	-0.827479	0.4085
RESID^2(-5)	-0.011256	0.053820	-0.209144	0.8344

R-squared	0.044853	Mean dependent var	0.007161
Adjusted R-squared	0.032884	S.D. dependent var	0.015446
S.E. of regression	0.015189	Akaike info criterion	-5.521722
Sum squared resid	0.092058	Schwarz criterion	-5.462405
Log likelihood	1124.149	Hannan-Quinn criter.	-5.498243
F-statistic	3.747387	Durbin-Watson stat	1.865295
Prob(F-statistic)	0.002507		

Heteroskedasticity Test: ARCH

F-statistic	3.102500	Prob. F(6,397)	0.0056
Obs*R-squared	18.09477	Prob. Chi-Square(6)	0.0060

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/26/26 Time: 12:58
Sample (adjusted): 1992M08 2026M03
Included observations: 404 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005386	0.001036	5.199077	0.0000
RESID^2(-1)	0.144140	0.053968	2.670842	0.0079
RESID^2(-2)	0.148451	0.054546	2.721585	0.0068
RESID^2(-3)	0.020590	0.055107	0.373627	0.7089
RESID^2(-4)	-0.045708	0.055108	-0.829429	0.4074
RESID^2(-5)	-0.012020	0.054545	-0.220370	0.8257
RESID^2(-6)	0.003764	0.053956	0.069757	0.9444

R-squared	0.044789	Mean dependent var	0.007170
Adjusted R-squared	0.030353	S.D. dependent var	0.015463
S.E. of regression	0.015227	Akaike info criterion	-5.514325
Sum squared resid	0.092048	Schwarz criterion	-5.444993
Log likelihood	1120.894	Hannan-Quinn criter.	-5.486879
F-statistic	3.102500	Durbin-Watson stat	1.865561
Prob(F-statistic)	0.005563		

Date: 04/26/26 Time: 12:48
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.150	0.150	9.2516	0.002
		2	0.148	0.128	18.279	0.000
		3	0.055	0.017	19.522	0.000
		4	-0.006	-0.036	19.538	0.001
		5	-0.004	-0.009	19.545	0.002
		6	-0.004	0.003	19.553	0.003
		7	0.045	0.052	20.413	0.005
		8	0.005	-0.007	20.424	0.009
		9	0.040	0.028	21.100	0.012
		10	0.023	0.011	21.324	0.019
		11	-0.000	-0.012	21.324	0.030
		12	-0.042	-0.049	22.086	0.037
		13	0.001	0.016	22.086	0.054
		14	-0.002	0.009	22.087	0.077
		15	-0.009	-0.008	22.122	0.105
		16	0.020	0.016	22.293	0.134

图 18: 序列 DF_LN_BRENT 的 MA(1) 模型条件异方差检验图

图 18显示：残差序列存在显著的条件异方差性。为此，本文进一步构建 GARCH 族模型。

4.3　GARCH 族模型构建与检验

序列 DF_LN_BRENT 的直方图与描述性统计如图 19所示。

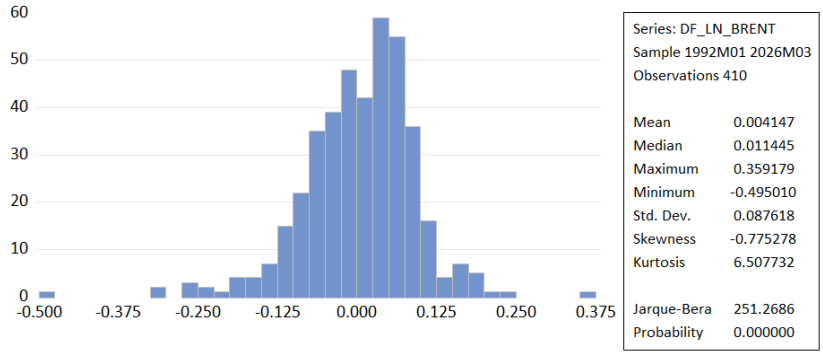


图 19: 序列 DF_LN_BRENT 的直方图与描述性统计图

由图 19可知，序列 DF_LN_BRENT 同样具备**尖峰厚尾**特征，故采用 t 分布构建 GARCH 族模型。将各 GARCH 族模型系数显著性检验和信息准则结果分别汇总到表 9和表 10中。

表 9: DF_LN_BRENT 序列 GARCH 族模型系数显著性检验结果

模型名称	常数项 C	ARCH 项	非对称项	GARCH 项
MA(1)-GARCH(1,0)-t	**0.0000**	**0.0020**	-	-
MA(1)-GARCH(2,0)-t	0.0000	0.0119 / 0.1624	-	-
MA(1)-GARCH(1,1)-t	0.0711	0.0106	-	0.0000
MA(1)-GARCH(1,2)-t	0.1173	0.0247	-	0.3410 / 0.6248
MA(1)-GARCH(2,1)-t	0.2584	0.0091 / 0.1120	-	0.0000
MA(1)-GARCH(2,2)-t	0.4571	0.0124 / 0.0557	-	0.0162 / 0.6028
MA(1)-TARCH(1,0)-t	0.0000	0.4087	0.0027	-
MA(1)-TARCH(2,0)-t	0.0000	0.5153 / 0.3085	0.0049	-
MA(1)-TARCH(1,1)-t	0.0000	0.6256	0.0044	0.4610
MA(1)-TARCH(1,2)-t	0.0160	0.5665	0.0077	0.8394 / 0.0148
MA(1)-TARCH(2,1)-t	0.0001	0.6992 / 0.3794	0.0056	0.7249
MA(1)-TARCH(2,2)-t	0.0060	0.5901 / 0.3148	0.0031	0.5210 / 0.3916
MA(1)-EGARCH(1,0)-t	**0.0000**	**0.0094**	**0.0081**	-
MA(1)-EGARCH(2,0)-t	**0.0000**	**0.0164 / 0.0096**	**0.0168**	-
MA(1)-EGARCH(1,1)-t	0.0310	0.0130	0.1428	0.0000
MA(1)-EGARCH(1,2)-t	0.0627	0.0067	0.0121	0.2432 / 0.0000
MA(1)-EGARCH(2,1)-t	0.0681	0.0246 / 0.6542	0.1683	0.0000
MA(1)-EGARCH(2,2)-t	0.0365	0.0107 / 0.1686	0.0384	0.7327 / 0.0000

表 10: DF_LN_BRENT 序列 GARCH 族模型信息准则结果对比

模型名称	MA(1) 系数	AIC	SC	HQ
MA(1)-GARCH(1,0)-t	0.187003	-2.2038	-2.1646	-2.1883
MA(1)-GARCH(2,0)-t	0.217883	-2.2071	-2.1581	-2.1877
MA(1)-GARCH(1,1)-t	0.227823	-2.2190	-2.1700	-2.1996
MA(1)-GARCH(1,2)-t	0.223801	-2.2153	-2.1566	-2.1921
MA(1)-GARCH(2,1)-t	0.213462	-2.2204	-2.1616	-2.1971
MA(1)-GARCH(2,2)-t	0.218747	-2.2171	-2.1485	-2.1900
MA(1)-TARCH(1,0)-t	0.230716	-2.2274	-2.1784	-2.2080
MA(1)-TARCH(2,0)-t	0.237428	-2.2263	-2.1676	-2.2031
MA(1)-TARCH(1,1)-t	0.236274	-2.2236	-2.1648	-2.2003
MA(1)-TARCH(1,2)-t	0.220951	-2.2232	-2.1546	-2.1961
MA(1)-TARCH(2,1)-t	0.237013	-2.2216	-2.1530	-2.1945
MA(1)-TARCH(2,2)-t	0.229717	-2.2236	-2.1452	-2.1926
MA(1)-EGARCH(1,0)-t	0.235877	-2.2009	-2.1520	-2.1816
MA(1)-EGARCH(2,0)-t	**0.243346**	**-2.2174**	**-2.1587**	**-2.1942**
MA(1)-EGARCH(1,1)-t	0.233722	-2.2256	-2.1668	-2.2024
MA(1)-EGARCH(1,2)-t	0.226163	-2.2277	-2.1591	-2.2005
MA(1)-EGARCH(2,1)-t	0.229811	-2.2212	-2.1526	-2.1941
MA(1)-EGARCH(2,2)-t	0.234614	-2.2297	-2.1513	-2.1987

由表 9和表 10，再结合显著性检验和信息准则可知，$\mathbf{MA\,(1)-EGARCH\,(2,0)-t}$ 为最优模型，所有参数均显著且信息准则表现最优，其结果如图 20所示。

Dependent Variable: DF_LN_BRENT
Method: ML ARCH - Student's t distribution (BFGS / Marquardt steps)
Date: 04/26/26 Time: 15:32
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments
Convergence achieved after 25 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1992M01
Presample variance: backcast (parameter = 0.7)
LOG(GARCH) = C(2) + C(3)*ABS(RESID(-1)/@SQRT(GARCH(-1))) + C(4)*ABS(RESID(-2)/@SQRT(GARCH(-2))) + C(5)*RESID(-1)/@SQRT(GARCH(-1))

Variable	Coefficient	Std. Error	z-Statistic	Prob.
MA(1)	0.243346	0.049593	4.906834	0.0000
Variance Equation				
C(2)	-5.634129	0.208354	-27.04120	0.0000
C(3)	0.348610	0.145203	2.400847	0.0164
C(4)	0.376054	0.145150	2.590787	0.0096
C(5)	-0.208818	0.087367	-2.390124	0.0168
T-DIST. DOF	10.93557	4.142522	2.639833	0.0083

R-squared	0.071245	Mean dependent var	0.004147
Adjusted R-squared	0.071245	S.D. dependent var	0.087618
S.E. of regression	0.084439	Akaike info criterion	-2.217445
Sum squared resid	2.916181	Schwarz criterion	-2.158672
Log likelihood	460.5763	Hannan-Quinn criter.	-2.194193
Durbin-Watson stat	1.867497		

Inverted MA Roots	-.24

图 20: MA(1)-EGARCH(2,0)-t 模型结果图

MA(1)-EGARCH(2,0)-t 模型的 ARCH-LM 检验 (滞后 4、5、6 阶) 和残差平方序列的 Q 检验结果如图 21所示。

Heteroskedasticity Test: ARCH

F-statistic	1.116352	Prob. F(4,401)	0.3483
Obs*R-squared	4.471297	Prob. Chi-Square(4)	0.3460

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/26/26 Time: 15:40
Sample (adjusted): 1992M06 2026M03
Included observations: 406 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.893152	0.145133	6.154010	0.0000
WGT_RESID^2(-1)	0.013600	0.060015	0.226604	0.8208
WGT_RESID^2(-2)	-0.007472	0.059421	-0.125751	0.9000
WGT_RESID^2(-3)	0.124660	0.059406	2.098452	0.0365
WGT_RESID^2(-4)	-0.001273	0.059983	-0.021217	0.9831

R-squared	0.011013	Mean dependent var	1.018035
Adjusted R-squared	0.001148	S.D. dependent var	1.923022
S.E. of regression	1.921918	Akaike info criterion	4.156763
Sum squared resid	1481.201	Schwarz criterion	4.206102
Log likelihood	-838.8228	Hannan-Quinn criter.	4.176290
F-statistic	1.116352	Durbin-Watson stat	1.693377
Prob(F-statistic)	0.348303		

Heteroskedasticity Test: ARCH

F-statistic	0.862251	Prob. F(6,397)	0.5227
Obs*R-squared	5.197003	Prob. Chi-Square(6)	0.5188

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/26/26 Time: 15:41
Sample (adjusted): 1992M08 2026M03
Included observations: 404 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.919390	0.161004	5.710347	0.0000
WGT_RESID^2(-1)	0.013849	0.060290	0.229708	0.8184
WGT_RESID^2(-2)	-0.009455	0.060278	-0.156857	0.8754
WGT_RESID^2(-3)	0.131356	0.060261	2.179767	0.0299
WGT_RESID^2(-4)	-0.002399	0.060263	-0.039807	0.9683
WGT_RESID^2(-5)	0.020165	0.060278	0.334524	0.7382
WGT_RESID^2(-6)	-0.048864	0.060266	-0.810801	0.4180

R-squared	0.012864	Mean dependent var	1.020413
Adjusted R-squared	-0.002055	S.D. dependent var	1.927477
S.E. of regression	1.929457	Akaike info criterion	4.169529
Sum squared resid	1477.953	Schwarz criterion	4.238860
Log likelihood	-835.2448	Hannan-Quinn criter.	4.196974
F-statistic	0.862251	Durbin-Watson stat	1.688886
Prob(F-statistic)	0.522717		

Heteroskedasticity Test: ARCH

F-statistic	0.904203	Prob. F(5,399)	0.4782
Obs*R-squared	4.537585	Prob. Chi-Square(5)	0.4749

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/26/26 Time: 15:41
Sample (adjusted): 1992M07 2026M03
Included observations: 405 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.879181	0.153435	5.730006	0.0000
WGT_RESID^2(-1)	0.012959	0.060174	0.215352	0.8296
WGT_RESID^2(-2)	-0.010029	0.060160	-0.166705	0.8677
WGT_RESID^2(-3)	0.124293	0.059562	2.086791	0.0375
WGT_RESID^2(-4)	-0.001929	0.060148	-0.032079	0.9744
WGT_RESID^2(-5)	0.019847	0.060155	0.329925	0.7416

R-squared	0.011204	Mean dependent var	1.019458
Adjusted R-squared	-0.001187	S.D. dependent var	1.925186
S.E. of regression	1.926328	Akaike info criterion	4.163813
Sum squared resid	1480.586	Schwarz criterion	4.223130
Log likelihood	-837.1721	Hannan-Quinn criter.	4.187291
F-statistic	0.904203	Durbin-Watson stat	1.691784
Prob(F-statistic)	0.478219		

Date: 04/26/26 Time: 15:42
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.010	0.010	0.0382	0.845
		2	-0.003	-0.004	0.0431	0.979
		3	0.087	0.088	3.2156	0.360
		4	0.002	0.000	3.2171	0.522
		5	0.013	0.014	3.2897	0.655
		6	-0.021	-0.029	3.4743	0.747
		7	0.048	0.049	4.4531	0.726
		8	-0.010	-0.014	4.4975	0.810
		9	0.052	0.058	5.6539	0.774
		10	0.087	0.078	8.8489	0.547
		11	-0.027	-0.025	9.1470	0.608
		12	-0.043	-0.054	9.9375	0.621
		13	0.015	0.004	10.037	0.691
		14	-0.007	-0.007	10.056	0.758
		15	0.011	0.022	10.110	0.813
		16	0.022	0.020	10.316	0.850

*Probabilities may not be valid for this equation specification.

图 21: MMA(1)-EGARCH(2,0)-t 模型的条件异方差检验结果图

根据图 21可知，各滞后期的 p 值均大于 0.05，表明残差序列不存在显著自相关。结合 ARCH-LM 检验结果可知，滞后 4、5、6 阶的 p 值均大于 0.05，这说明模型已充分捕捉原序列的条件异方差。

MA(1)-EGARCH(2,0)-t 模型的残差白噪声检验结果如图 22所示。

Date: 04/26/26 Time: 15:39
Sample (adjusted): 1992M02 2026M03
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.016	0.016	0.1085	
		2	0.003	0.003	0.1127	0.737
		3	-0.031	-0.031	0.5035	0.777
		4	-0.055	-0.054	1.7475	0.626
		5	0.026	0.028	2.0339	0.730
		6	-0.017	-0.018	2.1545	0.827
		7	0.009	0.006	2.1851	0.902
		8	-0.038	-0.040	2.8034	0.903
		9	-0.049	-0.046	3.7998	0.875
		10	0.047	0.047	4.7272	0.857
		11	0.051	0.050	5.8407	0.828
		12	-0.003	-0.013	5.8441	0.884
		13	-0.081	-0.082	8.6224	0.735
		14	-0.066	-0.055	10.458	0.656
		15	0.031	0.037	10.880	0.695
		16	-0.014	-0.020	10.960	0.755

*Probabilities may not be valid for this equation specification.

图 22: MA(1)-EGARCH(2,0)-t 模型的残差白噪声检验结果图

根据图 22可知，各滞后阶数的 p 值均大于 0.05，不拒绝残差序列为白噪声的原假设，表明模型已充分提取时间序列的动态信息。

4.4 模型方程与结果解读

由图 20可知，序列 DF_LN_BRENT 的 MA(1)-EGARCH(2,0)-t 模型方程如下：

$$\mathrm{DF_LN_BRENT}_t = \varepsilon_t + \theta\varepsilon_{t-1} = \varepsilon_t + 0.243346 \cdot \varepsilon_{t-1}, \tag{8}$$

$$\begin{aligned} \ln\left(\sigma_t^2\right) &= \omega + \alpha\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \gamma\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \beta\ln\left(\sigma_{t-1}^2\right) \\ &= -5.634129 + 0.348610 \cdot \left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| 0.376054 \cdot \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \\ &\quad - 0.208818 \cdot \ln\left(\sigma_{t-1}^2\right), \end{aligned} \tag{9}$$

其中，ε_t 为扰动项，σ_t^2 为条件方差，θ 为移动平均系数，ω 为方差方程常数项，α 为 ARCH 项系数，γ 为非对称项系数，β 为 GARCH 项系数，模型扰动项 $\varepsilon_t \sim t(10.9.557)$。

上述均值方程(9)和方差方程(9)表明：**布伦特原油收益率存在短期记忆性，外部冲击会加剧价格波动，波动具有强持续性，且存在显著的负向杠杆效应。**

5 构建序列 LN_DUBAI 的时间序列模型

5.1 平稳性与非白噪声检验

5.1.1 ADF 单位根检验

对序列 LN_DUBAI 做 ADF 检验得到三种情形的 p 值分别为：None(0.8489)、Constant(0.4904)、Constant,Linear Trend(0.1390) 均大于 0.05，故原序列存在单位根，不满足平稳性。对其进行一阶差分得到序列

$$\text{DF_LN_DUBAI} = \text{LN_DUBAI}_t - \text{LN_DUBAI}_{t-1}, \tag{10}$$

再对差分后的序列重新进行 ADF 检验，检验结果如表 11所示。

表 11: 序列 DF_LN_DUBAI 的 ADF 检验不同情形的信息准则对比

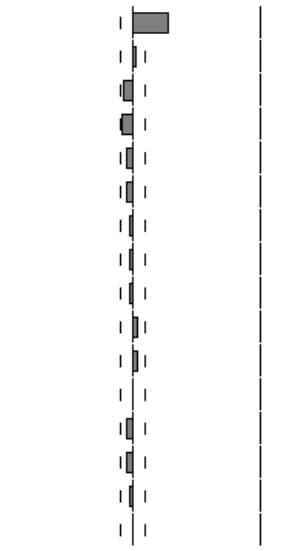
检验情形	p 值	AIC 准则	SC 准则	HQ 准则
None	0.0000	**-2.076381**	**-2.066567**	**-2.072498**
Constant	0.0000	-2.073570	-2.053943	-2.065804
Constant,Linear Trend	0.0000	-2.068842	-2.039402	-2.057194

由表可知，三种检验情形下 p 值均小于 0.05，拒绝存在单位根的原假设，序列平稳。根据 AIC/SC/HQ 信息准则，选择**无截距项、无时间趋势**的情形进行建模。

5.1.2 自相关与偏自相关函数分析

序列 DF_LN_DUBAI 自相关与偏自相关函数检验结果如图 23所示。

Date: 04/26/26 Time: 16:26
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments



Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.290	0.290	34.751	0.000
		2	0.027	-0.063	35.048	0.000
		3	-0.070	-0.066	37.102	0.000
		4	-0.083	-0.045	39.970	0.000
		5	-0.045	-0.010	40.825	0.000
		6	-0.045	-0.038	41.669	0.000
		7	-0.032	-0.019	42.090	0.000
		8	-0.025	-0.019	42.344	0.000
		9	-0.026	-0.023	42.628	0.000
		10	0.034	0.044	43.104	0.000
		11	0.037	0.008	43.669	0.000
		12	0.002	-0.021	43.671	0.000
		13	-0.057	-0.057	45.031	0.000
		14	-0.049	-0.012	46.040	0.000
		15	-0.019	-0.001	46.196	0.000
		16	0.000	-0.001	46.196	0.000

图 23: 序列 DF_LN_DUBAI 的 ACF 和 PACF 相关图

由图可知，序列 DF_LN_DUBAI 为非白噪声序列，其**自相关函数 1 阶截尾，偏自相关函数 1 阶截尾**。

5.2 ARMA 模型构建与条件异方差检验

5.2.1 模型构建与优选

根据序列 DF_LN_DUBAI 自相关与偏自相关函数的特征，对 DF_LN_DUBAI 构建多种 ARMA 模型，并将各个 ARMA 模型的系数显著性检验结果和 AIC/SC/HQ 值汇总到如下表 12中。

表 12: DF_LN_DUBAI 序列 ARMA 模型系数显著性检验和信息准则对比图

模型	AR(1)	AR(2)	MA(1)	MA(2)	AIC	SC	HQ
ARMA(1,1)	0.1009	-	0.1818	-	-2.072224	-2.042838	-2.060598
ARMA(2,2)	0.0000	0.0388	0.0000	0.6405	-2.072927	-2.023950	-2.053551
ARMA(2,1)	**0.0000**	**0.0000**	**0.0000**	-	-2.077152	-2.037970	-2.061651
ARMA(1,2)	0.8909	-	0.5020	0.6261	-2.068641	-2.029459	-2.053139
AR(1)	**0.0000**	-	-	-	-2.073362	-2.053771	-2.065611
AR(2)	0.0000	0.0769	-	-	-2.073740	-2.044354	-2.062114
MA(1)	-	-	**0.0000**	-	-2.073296	-2.053705	-2.065545
MA(2)	-	-	**0.0000**	**0.0418**	-2.073392	-2.044005	-2.061765

由表，再结合系数显著性与信息准则，最优均值模型为 **AR(1)**，模型结果如图 24所示。

Dependent Variable: DF_LN_DUBAI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 04/26/26 Time: 16:35
Sample: 1992M02 2026M03
Included observations: 410
Convergence achieved after 19 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.328258	0.034799	9.433034	0.0000
SIGMASQ	0.007290	0.000249	29.23189	0.0000

R-squared	0.092922	Mean dependent var	0.005169
Adjusted R-squared	0.090698	S.D. dependent var	0.089757
S.E. of regression	0.085590	Akaike info criterion	-2.073362
Sum squared resid	2.988837	Schwarz criterion	-2.053771
Log likelihood	427.0391	Hannan-Quinn criter.	-2.065611
Durbin-Watson stat	1.838644		

Inverted AR Roots	.33

图 24: 序列 DF_LN_DUBAI 的 AR(1) 模型结果图

5.2.2 适应性与条件异方差检验

序列 DF_LN_DUBAI 的 AR(1) 模型的适应性检验结果如图 25所示。

Date: 04/26/26　Time: 16:37
Sample (adjusted): 1992M02 2026M03
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.022	0.022	0.1960	
		2	-0.041	-0.042	0.9074	0.341
		3	-0.064	-0.062	2.6070	0.272
		4	-0.062	-0.062	4.2278	0.238
		5	-0.008	-0.011	4.2524	0.373
		6	-0.033	-0.042	4.6991	0.454
		7	-0.013	-0.020	4.7670	0.574
		8	-0.011	-0.020	4.8221	0.682
		9	-0.035	-0.043	5.3520	0.719
		10	0.045	0.038	6.1901	0.721
		11	0.028	0.019	6.5231	0.770
		12	0.006	0.001	6.5399	0.835
		13	-0.054	-0.054	7.7908	0.801
		14	-0.038	-0.031	8.4090	0.816
		15	-0.000	-0.003	8.4090	0.867
		16	0.011	0.004	8.4653	0.904

图 25: 序列 DF_LN_DUBAI 的 AR(1) 模型适应性检验图

图 25显示：AR(1) 模型残差通过白噪声检验，均值信息提取充分。

序列 DF_LN_DUBAI 的 AR(1) 模型的滞后 4、5、6 阶的 ARCH-LM 检验和残差平方序列的 Q 检验结果如图 26所示。

Heteroskedasticity Test: ARCH

F-statistic	3.796903	Prob. F(4,401)	0.0048
Obs*R-squared	14.81584	Prob. Chi-Square(4)	0.0051

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/26/26 Time: 17:37
Sample (adjusted): 1992M06 2026M03
Included observations: 406 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005654	0.001332	4.245810	0.0000
RESID^2(-1)	0.146861	0.079401	1.849602	0.0651
RESID^2(-2)	0.239668	0.080137	2.990723	0.0030
RESID^2(-3)	-0.006374	0.080133	-0.079537	0.9366
RESID^2(-4)	-0.119545	0.079397	-1.505655	0.1329

R-squared	0.036492	Mean dependent var	0.007349
Adjusted R-squared	0.026881	S.D. dependent var	0.021878
S.E. of regression	0.021582	Akaike info criterion	-4.821653
Sum squared resid	0.186783	Schwarz criterion	-4.772314
Log likelihood	983.7955	Hannan-Quinn criter.	-4.802125
F-statistic	3.796903	Durbin-Watson stat	1.394706
Prob(F-statistic)	0.004819		

Heteroskedasticity Test: ARCH

F-statistic	2.509997	Prob. F(6,397)	0.0214
Obs*R-squared	14.76541	Prob. Chi-Square(6)	0.0222

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/26/26 Time: 17:38
Sample (adjusted): 1992M08 2026M03
Included observations: 404 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005791	0.001440	4.022127	0.0001
RESID^2(-1)	0.145115	0.080233	1.808668	0.0713
RESID^2(-2)	0.238395	0.080925	2.945861	0.0034
RESID^2(-3)	-0.003193	0.083164	-0.038396	0.9694
RESID^2(-4)	-0.115022	0.083173	-1.382925	0.1675
RESID^2(-5)	-0.010772	0.080933	-0.133098	0.8942
RESID^2(-6)	-0.012467	0.080233	-0.155381	0.8766

R-squared	0.036548	Mean dependent var	0.007369
Adjusted R-squared	0.021987	S.D. dependent var	0.021931
S.E. of regression	0.021688	Akaike info criterion	-4.806930
Sum squared resid	0.186738	Schwarz criterion	-4.737599
Log likelihood	977.9999	Hannan-Quinn criter.	-4.779485
F-statistic	2.509997	Durbin-Watson stat	1.393192
Prob(F-statistic)	0.021411		

Heteroskedasticity Test: ARCH

F-statistic	3.024778	Prob. F(5,399)	0.0108
Obs*R-squared	14.79068	Prob. Chi-Square(5)	0.0113

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 04/26/26 Time: 17:38
Sample (adjusted): 1992M07 2026M03
Included observations: 405 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.005719	0.001385	4.128523	0.0000
RESID^2(-1)	0.145498	0.080014	1.818415	0.0698
RESID^2(-2)	0.239765	0.080339	2.984418	0.0030
RESID^2(-3)	-0.003382	0.082949	-0.040767	0.9675
RESID^2(-4)	-0.118030	0.080341	-1.469110	0.1426
RESID^2(-5)	-0.012237	0.080013	-0.152933	0.8785

R-squared	0.036520	Mean dependent var	0.007358
Adjusted R-squared	0.024447	S.D. dependent var	0.021905
S.E. of regression	0.021635	Akaike info criterion	-4.814291
Sum squared resid	0.186764	Schwarz criterion	-4.754974
Log likelihood	980.8940	Hannan-Quinn criter.	-4.790812
F-statistic	3.024778	Durbin-Watson stat	1.393504
Prob(F-statistic)	0.010799		

Date: 04/26/26 Time: 17:38
Sample (adjusted): 1992M02 2026M03
Included observations: 410 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.075	0.075	2.3118	0.128
		2	0.097	0.092	6.2454	0.044
		3	0.022	0.009	6.4475	0.092
		4	-0.020	-0.032	6.6145	0.158
		5	-0.008	-0.008	6.6430	0.249
		6	-0.022	-0.016	6.8461	0.335
		7	0.006	0.011	6.8603	0.444
		8	-0.008	-0.006	6.8881	0.549
		9	0.023	0.023	7.1037	0.626
		10	0.007	0.004	7.1252	0.714
		11	-0.008	-0.012	7.1492	0.787
		12	-0.019	-0.020	7.2994	0.837
		13	-0.001	0.005	7.2996	0.886
		14	0.005	0.009	7.3086	0.922
		15	-0.005	-0.005	7.3187	0.948
		16	0.005	0.003	7.3297	0.966

图 26: 序列 DF_LN_DUBAI 的 AR(1) 模型条件异方差检验图

图 26显示：残差序列存在显著的条件异方差性。为此，本文进一步构建 GARCH 族模型。

5.3　GARCH 族模型构建与检验

序列 DF_LN_DUBAI 的直方图与描述性统计如图 27所示。

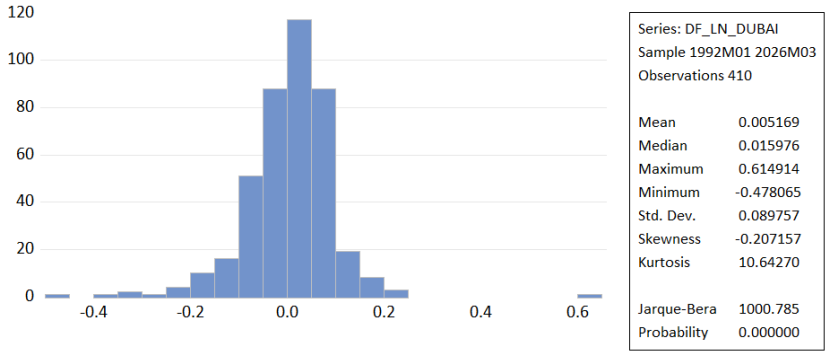


图 27: 序列 DF_LN_DUBAI 的直方图与描述性统计图

由图可知，序列 DF_LN_DUBAI 同样具备**尖峰厚尾**特征，故采用 t 分布构建 GARCH 族模型。将各 GARCH 族模型系数显著性检验和信息准则结果分别汇总到表 13和表 14中。

表 13: DF_LN_DUBAI 序列 GARCH 族模型系数显著性检验结果

模型名称	常数项 C	ARCH 项	非对称项	GARCH 项
AR(1)-GARCH(1,0)-t	0.0000	0.0876	-	-
AR(1)-GARCH(2,0)-t	0.0000	0.2283 / 0.2437	-	-
AR(1)-GARCH(1,1)-t	0.1577	0.3083	-	0.6335
AR(1)-GARCH(1,2)-t	0.0444	0.1967	-	0.1310 / 0.1013
AR(1)-GARCH(2,1)-t	0.6293	0.3454 / 0.9815	-	0.8787
AR(1)-GARCH(2,2)-t	0.0009	0.1269 / 0.1441	-	0.9455 / 0.2267
AR(1)-TARCH(1,0)-t	0.0000	0.2956	0.0063	-
AR(1)-TARCH(2,0)-t	0.0000	0.3888 / 0.5015	0.0076	-
AR(1)-TARCH(1,1)-t	0.0000	0.5026	0.0068	0.5418
AR(1)-TARCH(1,2)-t	0.0001	0.5900	0.0057	0.6664 / 0.7452
AR(1)-TARCH(2,1)-t	0.0000	0.5036 / 0.9671	0.0072	0.6278
AR(1)-TARCH(2,2)-t	0.0000	0.5998 / 0.0964	0.0065	0.2366 / 0.9086
AR(1)-EGARCH(1,0)-t	**0.0000**	**0.0013**	**0.0101**	-
AR(1)-EGARCH(2,0)-t	**0.0000**	**0.0015 / 0.0137**	**0.0166**	-
AR(1)-EGARCH(1,1)-t	**0.0118**	**0.0099**	**0.0218**	**0.0000**
AR(1)-EGARCH(1,2)-t	0.0205	0.0101	0.0171	0.0610 / 0.1884
AR(1)-EGARCH(2,1)-t	0.0338	0.0183 / 0.7554	0.0175	0.0003
AR(1)-EGARCH(2,2)-t	0.0390	0.0131 / 0.4136	0.0146	0.4972 / 0.0789

表 14: DF_LN_DUBAI 序列 GARCH 族模型信息准则结果对比

模型名称	AR(1) 系数 (p 值)	AIC	SC	HQ
AR(1)-GARCH(1,0)-t	0.0000	-2.150568	-2.111314	-2.135037
AR(1)-GARCH(2,0)-t	0.0001	-2.127236	-2.078168	-2.107822
AR(1)-GARCH(1,1)-t	0.0003	-2.044434	-1.995366	-2.025020
AR(1)-GARCH(1,2)-t	0.0000	-2.085860	-2.026979	-2.062563
AR(1)-GARCH(2,1)-t	0.0018	-2.046218	-1.987337	-2.022921
AR(1)-GARCH(2,2)-t	0.0000	-2.193008	-2.124314	-2.165828
AR(1)-TARCH(1,0)-t	0.0000	-2.310822	-2.261755	-2.291408
AR(1)-TARCH(2,0)-t	0.0000	-2.308010	-2.249129	-2.284713
AR(1)-TARCH(1,1)-t	0.0000	-2.308616	-2.249735	-2.285319
AR(1)-TARCH(1,2)-t	0.0000	-2.304437	-2.235743	-2.277257
AR(1)-TARCH(2,1)-t	0.0000	-2.303728	-2.235034	-2.276548
AR(1)-TARCH(2,2)-t	0.0000	-2.304617	-2.226109	-2.273554
AR(1)-EGARCH(1,0)-t	0.0000	-2.273970	-2.224903	-2.254556
AR(1)-EGARCH(2,0)-t	0.0000	-2.284499	-2.225618	-2.261202
AR(1)-EGARCH(1,1)-t	0.0000	**-2.296263**	**-2.237382**	**-2.272965**
AR(1)-EGARCH(1,2)-t	0.0000	-2.294569	-2.225875	-2.267389
AR(1)-EGARCH(2,1)-t	0.0000	-2.291505	-2.222811	-2.264325
AR(1)-EGARCH(2,2)-t	0.0000	-2.292254	-2.213746	-2.261191

由表，再结合显著性检验和信息准则可知，$\mathbf{MA(1)-EGARCH(1,1)-t}$ 为最优模型，所有参数均显著且信息准则表现最优，其结果如图 28所示。

Dependent Variable: DF_LN_DUBAI
Method: ML ARCH - Student's t distribution (OPG - BHHH / Marquardt steps)
Date: 04/26/26 Time: 18:08
Sample (adjusted): 1992M03 2026M03
Included observations: 409 after adjustments
Convergence achieved after 49 iterations
Coefficient covariance computed using outer product of gradients
Presample variance: backcast (parameter = 0.7)
LOG(GARCH) = C(2) + C(3)*ABS(RESID(-1)/@SQRT(GARCH(-1))) + C(4)*RESID(-1)/@SQRT(GARCH(-1)) + C(5)*LOG(GARCH(-1))

Variable	Coefficient	Std. Error	z-Statistic	Prob.
AR(1)	0.261960	0.050929	5.143582	0.0000
Variance Equation				
C(2)	-1.782555	0.707882	-2.518153	0.0118
C(3)	0.381334	0.147765	2.580674	0.0099
C(4)	-0.198862	0.086718	-2.293206	0.0218
C(5)	0.707304	0.128915	5.486613	0.0000
T-DIST. DOF	6.870946	1.226171	5.603576	0.0000

R-squared	0.089001	Mean dependent var	0.005102
Adjusted R-squared	0.089001	S.D. dependent var	0.089856
S.E. of regression	0.085765	Akaike info criterion	-2.296263
Sum squared resid	3.001071	Schwarz criterion	-2.237382
Log likelihood	475.5857	Hannan-Quinn criter.	-2.272965
Durbin-Watson stat	1.727307		

Inverted AR Roots	.26

图 28: MA(1)-EGARCH(1,1)-t 模型结果图

MA(1)-EGARCH(1,1)-t 模型的 ARCH-LM 检验 (滞后 4、5、6 阶) 和残差平方序列的 Q 检验结果如图 29所示。

Heteroskedasticity Test: ARCH

F-statistic	0.248146	Prob. F(4,400)	0.9107
Obs*R-squared	1.002505	Prob. Chi-Square(4)	0.9094

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/26/26 Time: 18:10
Sample (adjusted): 1992M07 2026M03
Included observations: 405 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.250493	0.293013	4.267706	0.0000
WGT_RESID^2(-1)	0.024840	0.112191	0.221404	0.8249
WGT_RESID^2(-2)	-0.045907	0.112206	-0.409135	0.6827
WGT_RESID^2(-3)	-0.034833	0.112232	-0.310369	0.7564
WGT_RESID^2(-4)	-0.092916	0.112211	-0.828050	0.4081

R-squared	0.002475	Mean dependent var	1.112150
Adjusted R-squared	-0.007500	S.D. dependent var	4.095377
S.E. of regression	4.110705	Akaike info criterion	5.677335
Sum squared resid	6759.160	Schwarz criterion	5.726766
Log likelihood	-1144.660	Hannan-Quinn criter.	5.696901
F-statistic	0.248146	Durbin-Watson stat	1.200396
Prob(F-statistic)	0.910730		

Heteroskedasticity Test: ARCH

F-statistic	0.212895	Prob. F(5,398)	0.9570
Obs*R-squared	1.077643	Prob. Chi-Square(5)	0.9561

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/26/26 Time: 18:10
Sample (adjusted): 1992M08 2026M03
Included observations: 404 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.222887	0.314053	3.893880	0.0001
WGT_RESID^2(-1)	0.026278	0.112571	0.233434	0.8155
WGT_RESID^2(-2)	-0.046109	0.112480	-0.409928	0.6821
WGT_RESID^2(-3)	-0.034730	0.112503	-0.308708	0.7577
WGT_RESID^2(-4)	-0.093016	0.112520	-0.826662	0.4089
WGT_RESID^2(-5)	0.029874	0.112620	0.265262	0.7909

R-squared	0.002667	Mean dependent var	1.113295
Adjusted R-squared	-0.009862	S.D. dependent var	4.100390
S.E. of regression	4.120559	Akaike info criterion	5.684595
Sum squared resid	6757.644	Schwarz criterion	5.744022
Log likelihood	-1142.288	Hannan-Quinn criter.	5.708119
F-statistic	0.212895	Durbin-Watson stat	1.200656
Prob(F-statistic)	0.956978		

Heteroskedasticity Test: ARCH

F-statistic	0.324053	Prob. F(6,396)	0.9243
Obs*R-squared	1.969019	Prob. Chi-Square(6)	0.9225

Test Equation:
Dependent Variable: WGT_RESID^2
Method: Least Squares
Date: 04/26/26 Time: 18:11
Sample (adjusted): 1992M09 2026M03
Included observations: 403 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.330633	0.333865	3.985540	0.0001
WGT_RESID^2(-1)	0.025159	0.112731	0.223179	0.8235
WGT_RESID^2(-2)	-0.050572	0.112745	-0.448557	0.6540
WGT_RESID^2(-3)	-0.034753	0.112660	-0.308475	0.7579
WGT_RESID^2(-4)	-0.092953	0.112672	-0.824987	0.4099
WGT_RESID^2(-5)	0.028266	0.112813	0.250553	0.8023
WGT_RESID^2(-6)	-0.106318	0.112797	-0.942565	0.3465

R-squared	0.004886	Mean dependent var	1.115582
Adjusted R-squared	-0.010192	S.D. dependent var	4.105228
S.E. of regression	4.126095	Akaike info criterion	5.689757
Sum squared resid	6741.765	Schwarz criterion	5.759217
Log likelihood	-1139.486	Hannan-Quinn criter.	5.717256
F-statistic	0.324053	Durbin-Watson stat	1.201378
Prob(F-statistic)	0.924299		

Date: 04/26/26 Time: 18:11
Sample (adjusted): 1992M03 2026M03
Included observations: 409 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.005	0.005	0.0100	0.920
		2	-0.009	-0.009	0.0436	0.978
		3	-0.007	-0.007	0.0623	0.996
		4	-0.018	-0.018	0.2023	0.995
		5	0.006	0.006	0.2163	0.999
		6	-0.021	-0.021	0.3962	0.999
		7	0.004	0.004	0.4039	1.000
		8	-0.001	-0.002	0.4043	1.000
		9	0.010	0.010	0.4437	1.000
		10	0.016	0.015	0.5496	1.000
		11	-0.013	-0.013	0.6234	1.000
		12	-0.015	-0.015	0.7179	1.000
		13	0.003	0.003	0.7208	1.000
		14	0.003	0.003	0.7256	1.000
		15	0.017	0.016	0.8447	1.000
		16	0.004	0.004	0.8512	1.000

*Probabilities may not be valid for this equation specification.

图 29: MA(1)-EGARCH(1,1)-t 模型的条件异方差检验结果图

根据图可知，各滞后期的 p 值均大于 0.05，表明残差序列不存在显著自相关。结合 ARCH-LM 检验结果可知，滞后 4、5、6 阶的 p 值均大于 0.05，这说明模型已充分捕捉原序列的条件异方差。

MA(1)-EGARCH(1,1)-t 模型的残差白噪声检验结果如图 30所示。

Date: 04/26/26 Time: 18:09
Sample (adjusted): 1992M03 2026M03
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.055	0.055	1.2435	
		2	-0.013	-0.016	1.3158	0.251
		3	-0.049	-0.047	2.2937	0.318
		4	-0.042	-0.037	3.0246	0.388
		5	-0.023	-0.021	3.2517	0.517
		6	-0.020	-0.021	3.4118	0.637
		7	-0.007	-0.010	3.4344	0.753
		8	-0.015	-0.019	3.5340	0.832
		9	-0.032	-0.034	3.9619	0.861
		10	0.037	0.037	4.5312	0.873
		11	0.057	0.049	5.8936	0.824
		12	-0.003	-0.013	5.8971	0.880
		13	-0.061	-0.059	7.4552	0.826
		14	-0.042	-0.031	8.2199	0.829
		15	-0.004	0.002	8.2264	0.877
		16	-0.009	-0.013	8.2580	0.913

*Probabilities may not be valid for this equation specification.

图 30: MA(1)-EGARCH(1,1)-t 模型的残差白噪声检验结果图

根据图可知，各滞后阶数的 p 值均大于 0.05，不拒绝残差序列为白噪声的原假设，表明模型已充分提取时间序列的动态信息。

5.4 模型方程与结果解读

由图 28可知，序列 DF_LN_DUBAI 的 MA(1)-EGARCH(1,1)-t 模型方程如下：

$$\text{DF_LN_DUBAI}_t = \varepsilon_t + \theta\varepsilon_{t-1} = \varepsilon_t + 0.261960 \cdot \varepsilon_{t-1}, \tag{11}$$

$$\begin{aligned} \ln\left(\sigma_t^2\right) &= \omega + \alpha\left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + \gamma\frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \beta\ln\left(\sigma_{t-1}^2\right) \\ &= -1.782555 + 0.381334 \cdot \left|\frac{\varepsilon_{t-1}}{\sigma_{t-1}}\right| + -0.198862 \cdot \frac{\varepsilon_{t-1}}{\sigma_{t-1}} \\ &\quad + 0.707304 \cdot \ln\left(\sigma_{t-1}^2\right), \end{aligned} \tag{12}$$

其中，ε_t 为扰动项，σ_t^2 为条件方差，θ 为移动平均系数，ω 为方差方程常数项，α 为 ARCH 项系数，γ 为非对称项系数，β 为 GARCH 项系数，模型扰动项 $\varepsilon_t \sim t(6.870946)$。

上述均值方程和方差方程表明：**迪拜原油收益率存在短期记忆性，外部冲击会加剧价格波动，波动具有强持续性，且存在显著的负向杠杆效应。**

6 三大原油价格波动特征对比分析

- **波动共性**：三大国际原油价格均为一阶单整序列，最优均值模型均为 MA(1)，最优波动模型均为 MA(1)-EGARCH(1,1)-t；均存在短期记忆性、波动聚集性与负向杠杆效应，符合金融时间序列典型特征。
- **区域差异**：Brent 作为全球基准，杠杆效应最显著；WTI 受北美市场影响，短期波动敏感性最强；Dubai 作为亚太基准，波动持续性最强，体现了不同定价区域的市场结构差异。

7 结论

本文基于 WTI、Brent 和 Dubai 三大国际原油 1992 年 1 月至 2026 年 3 月的月度价格数据，采用 ARMA-EGARCH 模型系统分析了其对数收益率序列的波动特征。主要结论如下：

- 三大原油价格均为非平稳序列，一阶差分后平稳且为白噪声；最优均值模型分别为 WTI 和 Brent 为 MA(1)，Dubai 为 AR(1)，表明原油市场具有短期可预测性。
- 所有序列均存在显著的条件异方差和尖峰厚尾分布，t 分布 GARCH 族模型能有效刻画波动聚集性。
- 三大市场均存在显著的杠杆效应，负向冲击对价格波动的放大作用强于正向冲击，其中 Brent 的杠杆效应最强。
- 波动持续性方面，WTI 最高 ($\beta = 0.7972$)，Brent 无 GARCH 项，冲击衰减最快；Dubai 的 ARCH 项最大，对即期信息反应最敏感。这些差异反映了三大区域市场结构、交易机制和定价权本质的不同。

本研究为国际原油市场的风险管理、衍生品定价及政策调控提供了量化依据，同时也验证了 EGARCH-t 模型在能源商品市场波动建模中的适用性。